

## Micro I, class 16

I specialized to money outcomes, but replaced the finite outcome set with  $X = \mathbb{R}_+$ . The set  $\mathcal{L}$  is now the set of *cumulative distribution functions* with bounded support (in particular for every  $F \in \mathcal{L}$ ,  $F(0) = 0$  and there is a number  $z_F > 0$  with  $F(z_F) = 1$ ). Let  $\delta_y$  be the cdf that puts probability 1 on the point  $y \geq 0$ . We defined  $\succsim \subset \mathcal{L} \times \mathcal{L}$  to be *risk averse* if

$$\delta_{\int z dF} \succsim F \quad \text{for all } F \in \mathcal{L}.$$

For expected-utility preferences, it is equivalent to

$$u\left(\int z dF\right) \geq \int u(z) dF \quad \text{for all } F \in \mathcal{L},$$

which in turn is equivalent to concavity of the vN-M utility (a result known as Jensen's inequality).<sup>1</sup>

I then introduced a simple two-asset portfolio problem:

$$\max_{\alpha \geq 0} \int u(w(1+r) + \alpha(x-r)) dF(x),$$

where  $\alpha$  is money invested in a risky asset,  $x$  is its rate of return,  $r$  is the rate of return on a safe asset, and  $F$  is a c.d.f. describing the distribution of the risky asset's rate of return (a distribution with bounded support in  $[-1, \infty)$ ). Investment in the safe asset is  $m = w - \alpha$ , where  $w$  is initial wealth. We concluded that, if  $\int x dF(x) > r$ —the expected rate of return on the risky asset is larger than the rate of return on the safe asset—, then (generically) *no one* with monotone, risk-averse, EU preferences invests zero in the risky asset. (The qualifier “generically” is because every concave function on the real line is differentiable at all but a countable number of points. A concave function on the real line might not be differentiable at isolated points, and the conclusion fails there, but these points are negligible.)

The next questions were these: what is a sensible definition of one person to be *more risk averse than* another; and what does a sensible definition imply for expected utility preferences? To preview I will suppose that one person is more risk averse than another if the willingness to pay of the first person to get rid of the risk in a distribution is always at least as large as the willingness to pay for the second to get rid of the risk: the *risk premium* of the first is always at least as large as the risk premium for the second. It turns out that, for expected utility preferences with a  $C^2$  vN-M utility with positive first derivative, it is equivalent to

$$r_1(z) = -\frac{u_1''(z)}{u_1'(z)} \geq -\frac{u_0''(z)}{u_0'(z)} = r_0(z) \quad \text{for all } z > 0.$$

The number  $r_i(z)$  is person  $i$ 's *Arrow-Pratt risk aversion measure* at  $z$ , a measure of local concavity of a function at a point. I will prove next class the Arrow-Pratt theorem on comparative risk aversion which asserts that several definitions of “more risk averse than” are equivalent. An implication we will also prove is that the demand for a risky asset (in our two-asset portfolio problem with one safe and one risky asset) *does not increase* with an increase in risk aversion. This result might not seem very suspenseful, but seemingly innocuous extensions of this intuitive result fail.

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<sup>1</sup>A proof is beyond our scope, but here are two possible approaches. One starts with extending the two-point definition of concavity to  $n$  points, as I sketched in our EU representation theorem. Then one can show that any c.d.f.  $F$  can be well-approximated by a finite-support distribution and taking limits. (The key concept here is *weak convergence* of distributions.) Another uses the fact that every concave function has a nonempty *supergradient* at every point, then integrating across the resulting supergradient inequality.